DATA SATURDAYS

#77 PARMA 2025

Data Virtualization con Azure SQL Database

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Bio

- Gianluca Hotz | @glhotz | ghotz@ugiss.org
- Consulente Indipendente
  - 30 anni con SQL Server (dalla versione 4.21 nel 1996)
  - «Database modeling & development, sizing & administration, modernization (upgrades & migrations), performance tuning, security»
- YNAP SQL Server DBA Team (part-time)
- «Community»
  - 25+ anni Microsoft MVP SQL Server/Data Platform (dal 1998)
  - «VMware Experts SQL Server»
  - Fondatore e presidente UGISS (ex «PASS Chapter»)

Systems requiring Data Movement

- DSS/OLAP/DWH
  - ETL (Extract, Transform, Load)
  - ELT (Extract, Load, Transform)
- IoT Data
- Master Data Management

Data Movement problems

- Costs
  - Duplicate storage costs
  - Effort to build/maintain data pipelines
- Velocity
  - Time to integrate new data → opportunities are missed
  - Data availability latency → opportunities are missed
- Security
  - Larger attack surface area
  - Potential inconsistency in security policies
- Quality
  - Meant to improve data quality, can still introduce new issues…
- Compliance
  - Greater governance issues

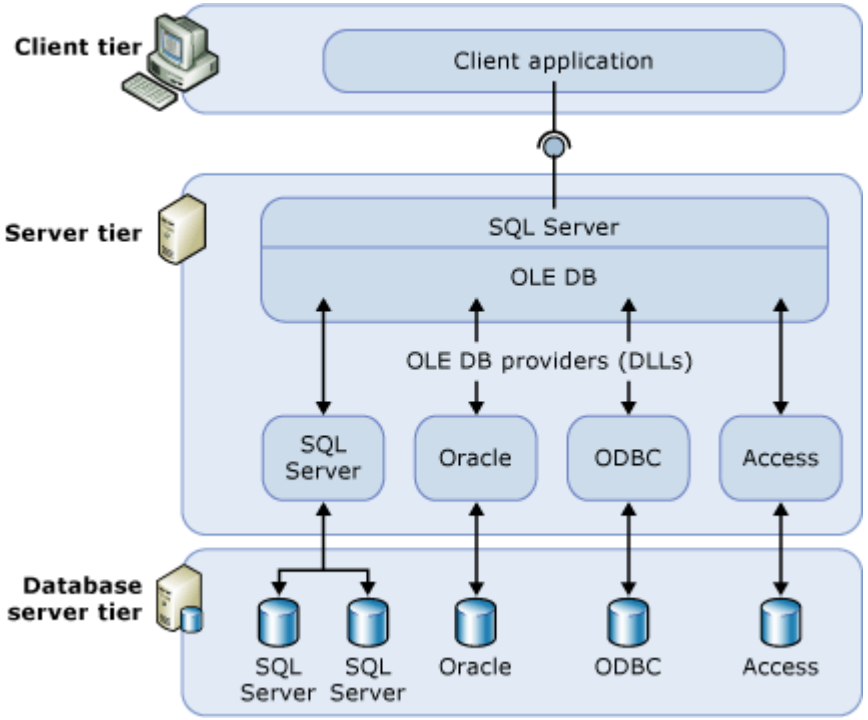
Data Virtualization principles

- Definition

  Unlike the traditional extract, transform, load (ETL) process, the data remains in place, and real-time access is given to the source system for the data. This reduces the risk of data errors, of the workload moving data around that may never be used, and it does not attempt to impose a single data model on the data https://en.wikipedia.org/wiki/Data_virtualization
- Data Abstraction
  - Hides complexities of data access, formats and structures
- Zero replication
  - No need to collect/copy data like a traditional ETL for transformations
- Real-time data
  - Always the latest available data

Microsoft Data Virtualization (past)

- Open Data Service Gateway
  - < SQL Server 7.0
- Linked Servers
  - Components
    - OLE DB provider
    - OLE DB data source
  - Usage
    - Access external data sources
    - Distributed queries/transactions
  - Support
    - SQL Server
    - Azure SQL Database Managed Instance



Microsoft Data Virtualization (today)

- SQL Server
  - Linked Servers
  - PolyBase
- Azure SQL Database
  - Data Virtualization (Preview)
  - Elastic Query (Preview)
- Azure SQL Managed Instance
  - Linked Servers
  - Data Virtualization
- Azure Synapse Analytics/Microsoft Fabric
  - Different native flavors of Data Virtualization

Benefits of MS Data Virtualization

- One source for all data with no data movement
  - Access directly from current location and single data hub to maintain
- T-SQL Language
  - Powerful and familiar language among database professionals
  - Same T-SQL surface across all platforms
- Security
  - Credential management and granular permissions control
- Cost flexibility
  - Available in all editions/service tiers

https://learn.microsoft.com/en-us/azure/azure-sql/database/data-virtualization-overview

Azure SQL Database DV (Preview)

- Execute queries with familiar T-SQL
  - on files storing data in common format
    - CSV, Parquet or Delta (1.0)
    - JSON indirectly (single row CSVs parsed with JSON_VALUE/OPENJSON)
  - against cloud object storages
    - Azure Data Lake Storage Gen2 or Azure Blob Storage
  - joining results with local relational data
  - in a secure and controlled way
    - Shared access signature (SAS), Managed Identity or Entra pass-through authentication via User Identity

Azure SQL Database DV T-SQL

- T-SQL support to define targets and authentication
  - CREATE DATABASE SCOPED CREDENTIAL
  - CREATE EXTERNAL DATA SOURCE
- T-SQL support
  - for ad-hoc querying (schema inferred at runtime)
    - OPENROWSET
  - for repetitive querying (schema defined statically)
    - CREATE EXTERNAL TABLE (CET)

Demo

- Azure SQL Database Data Virtualization

Elastic Query (Preview)

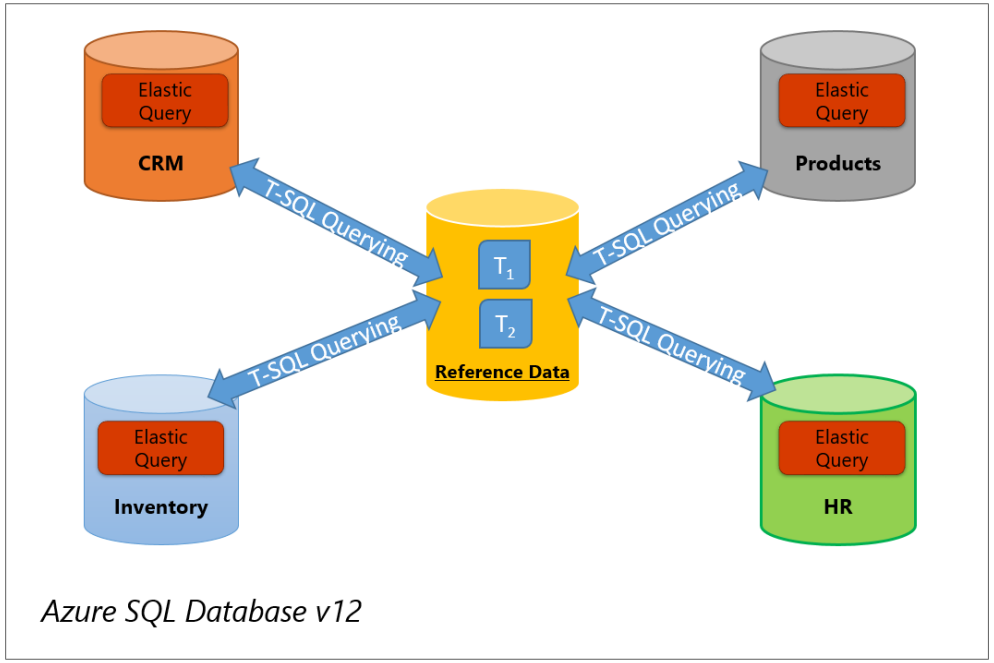
- Elastic Database Query
  - Vertical Partitioning
  - Horizontal Partitioning
- Elastic Database Client Library

- Azure SQL Database only…
- Still in preview after many years….

https://learn.microsoft.com/en-us/azure/azure-sql/database/elastic-query-overview

Elastic Query: vertical partitioning

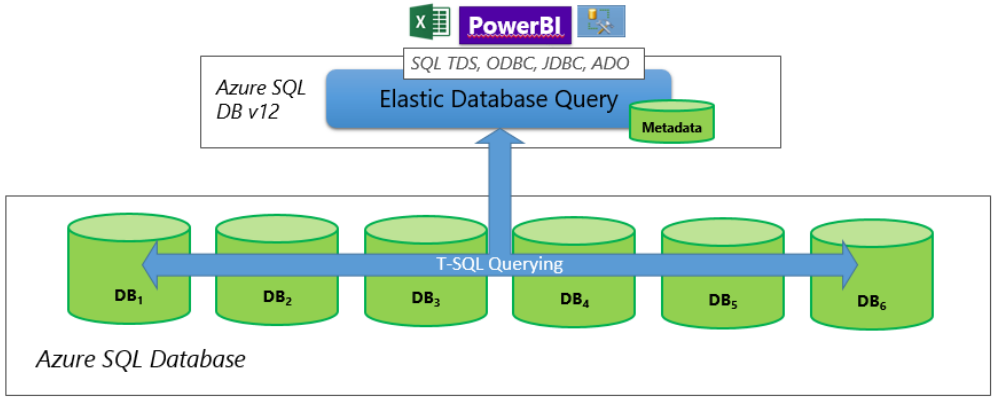
- Cross-database queries
  - Kind of Data Virtualization
    - Still Azure SQL Database only
- T-SQL support
  - CREATE DATABASE SCOPED CREDENTIAL
  - CREATE EXTERNAL DATA SOURCE
    - TYPE = RDBMS
  - CREATE EXTERNAL TABLE



https://learn.microsoft.com/en-us/azure/azure-sql/database/elastic-query-overview

Elastic Query: horizontal partitioning

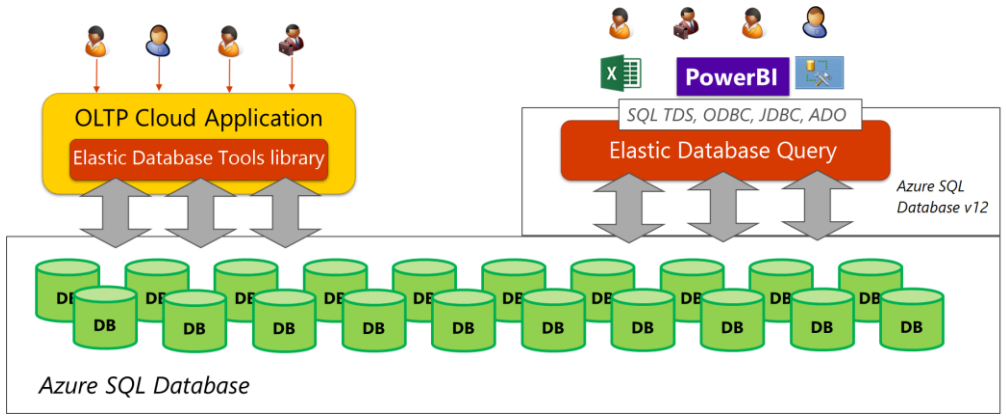
- Sharding
  - Scale-out (same schema)
  - Not really Data Virtualization
- T-SQL support
  - CREATE DATABASE SCOPED CREDENTIAL
  - CREATE EXTERNAL DATA SOURCE
    - TYPE = SHARD_MAP_MANAGER
  - CREATE EXTERNAL TABLE



https://learn.microsoft.com/en-us/azure/azure-sql/database/elastic-query-getting-started

Elastic Client Library

- Sharding
  - Scale-out (same schema)
  - Not really Data Virtualization
- No T-SQL support
  - Client-side support
  - UNION ALL semantic



https://learn.microsoft.com/en-us/azure/azure-sql/database/elastic-database-client-library

Azure SQL Database MI DV

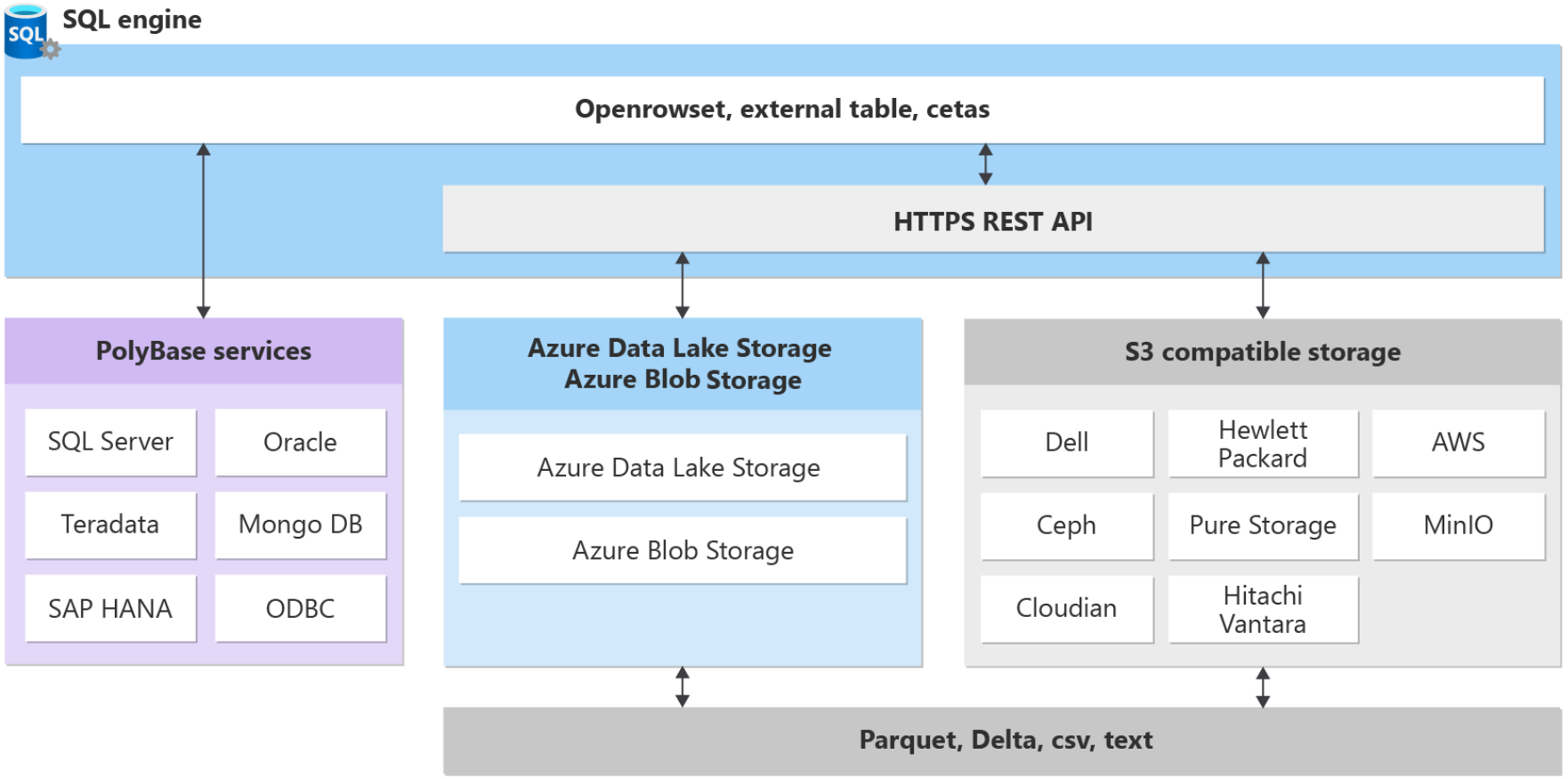
- Same core functionalities as Azure SQL Database DV
- Manual Statistics Creation
  - OPENROWSET
  - EXTERNAL TABLES
- Export data from into an external storage account
  - CREATE EXTERNAL TABLE AS SELECT (CETAS)

https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/data-virtualization-overview

Demo

- Azure SQL Database Managed Instance Data Virtualization

Data Virtualization support in 2025

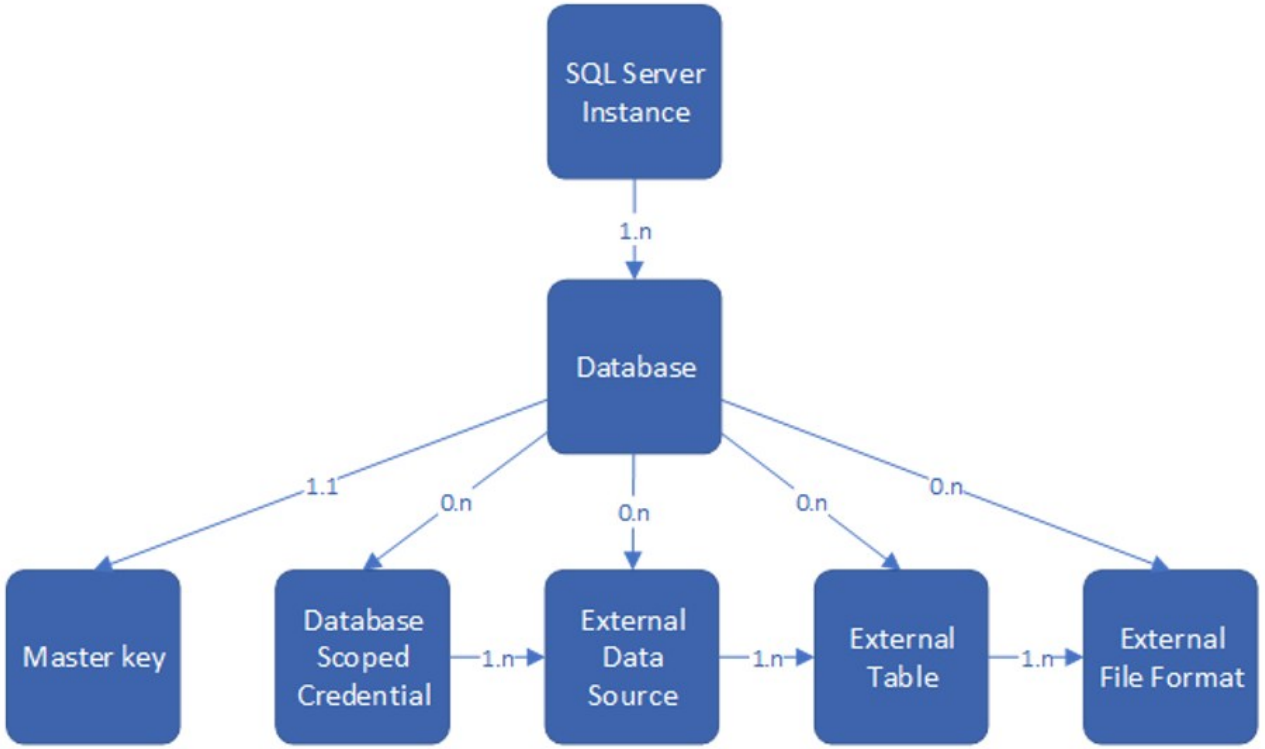


PolyBase support across versions

SQL Server 2016	SQL Server 2017	SQL Server 2019	SQL Server 2022	SQL Server 2025

• Hadoop
• Azure Blob Storage
	• OPENROWSET enhancements
	• CSV for Azure Blob Storage
	• Database Scoped Credential
		• SQL Server
		• Oracle
		• Azure Cosmos DB
		• MongoDB
		• Teradata
		• Linux support
		• Generic ODBC
			• CETAS
			• REST based framework
			  • Azure Blob
			  • ADLS
			  • S3
			  • CSV
			  • Parquet
			  • Delta
				• REST based framework in core engine

PolyBase Objects



PolyBase Implementation

- PolyBase services
  - SQL Server PolyBase Engine
  - SQL Server PolyBase Data Movement
- PolyBase REST API feature

Data source	PolyBase services	PolyBase REST API feature
Azure Blob Storage	No	Yes
Azure Data Lake Storage	No	Yes
S3-compatible object storage	No	Yes
SQL Server	Yes	No
Oracle	Yes	No
Teradata	Yes	No
MongoDB or Azure Cosmos DB API for MongoDB	Yes	No
Generic Open Database Connectivity (ODBC)	Yes	No
Bulk operations	Yes	No

PolyBase Support

- File formats
  - CSV
  - Parquet
  - Delimited text
  - Delta (read-only)

- OPENROWSET
  - Ad-hoc external data access
- CET
  - Create External Table
  - Reusable external data pointer
  - Supports statistics
- CETAS
  - Create External Table As Select
    - Exports manipulated data
    - CET target the exported data

Grazie!